🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 · STANDARD 6.AT.7

Simplify Algebraic Expressions

Lesson 6-15 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can simplify algebraic expressions by combining like terms.
Language Objective: I can explain my steps using the words like terms, combine, simplify, and coefficient.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Simplify Algebraic Expressions | Like Terms | Combine | Simplify | Coefficient | Term | Distributive property

👆 *Tap any word to see what it means and a picture.*

1. Combine __________ __________ to write a shorter expression with the same value.

2. Terms with the same variable part, like 3y and 8y, are __________.

3. To add or subtract like terms into a single term is to __________ them.

4. To write an expression in its shortest equal form is to __________ it.

5. The number multiplied by a variable is the __________.

6. A number or variable separated by + or – signs is a ______.

7. Multiplying each part of a sum by a number uses the ______ property.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

Simplify the total order expression. What does each part of the simplified expression tell you?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Simplify Algebraic Expressions** — Rewrite an expression in a shorter equivalent form by combining like terms and using properties.
  Simplificar expresiones algebraicas — Reescribir una expresión en una forma equivalente más corta combinando términos semejantes y usando propiedades.
- [ ] **Like Terms** — Terms with the same letter raised to the same power, like 2x and 5x.
  Términos semejantes — Términos con la misma letra elevada a la misma potencia, como 2x y 5x.
- [ ] **Combine** — To add or subtract terms with the same letter into one.
  Combinar — Sumar o restar términos con la misma letra en uno solo.
- [ ] **Simplify** — To write an expression in its shortest form.
  Simplificar — Escribir una expresión en su forma más corta.
- [ ] **Coefficient** — The number in front of a letter, like the 3 in 3x.
  Coeficiente — El número frente a una letra, como el 3 en 3x.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

The total order simplifies to ____m + ____. The ____ tells us ____ and the ____ tells us ____.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: You can only combine items of the same type, just like like terms.

Evidence: Students explain microphones combine with microphones ($4 + 2 + 6 = 12$) but not with stands, connecting this to combining like terms in algebra.

Reasoning: This evidence connects the definition of simplify algebraic expressions to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The total order simplifies to ____m + ____. The ____ tells us ____ and the ____ tells us ____.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Simplify Algebraic Expressions
2. **Notes 2:** Like Terms
3. **Notes 3:** Combine
4. **Notes 4:** Simplify
5. **Notes 5:** Coefficient
6. **Notes 6:** Term
7. **Notes 7:** Distributive property
8. **Watch for this mistake:** *A common mistake in Simplify Algebraic Expressions is sweeping a constant into the variable terms — for example, simplifying 6x + 4 + 2x as 12x by adding all three numbers (6 + 4 + 2 = 12) instead of recognizing that 4 has no variable and cannot combine with 6x and 2x. The correct simplified expression is 8x + 4 (combine 6x + 2x = 8x, and let the 4 stay on its own). Before you submit, check that every term you combined has the exact same variable part.*
9. **Try It 1:** A. 4y — *Subtract the coefficients of the like terms: 7 − 3 = 4. Keep the variable y, giving 4y.*
10. **Try It 2:** A. 9m + 4 — *6m + 3m = 9m. The constant 4 has no like term, so it stays. Result: 9m + 4.*